Zepto E-commerce SQL Data Analyst Portfolio Project

A complete, real-world data analyst portfolio project based on an e-commerce inventory dataset scraped from Zepto – one of India's fastest-growing quick-commerce startups.

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Who This Project Is For

Data Analyst Aspirants

Build a strong portfolio project for interviews and LinkedIn

SQL Learners

Hands-on practice with real-world e-commerce data

Interview Prep

Prepare for retail, e-commerce, or product analytics interviews

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Project Overview

Simulate how actual data analysts in e-commerce work behind the scenes using SQL to explore, clean, and derive business insights from real-world inventory data.

01

Database Setup

Create messy, real-world e-commerce inventory database

02

Exploratory Data Analysis

Explore product categories, availability, and pricing inconsistencies

03

Data Cleaning

Handle null values, remove invalid entries, convert pricing from paise to rupees

04

Business Insights

Write SQL queries around pricing, inventory, stock availability, and revenue

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Dataset Overview

The dataset was sourced from Kaggle and originally scraped from Zepto's official product listings. Each row represents a unique SKU (Stock Keeping Unit) for a product.

Duplicate product names exist because the same product may appear multiple times in different package sizes, weights, discounts, or categories – exactly how real catalog data looks.

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Key Data Columns

1

sku_id

Unique identifier for each product entry (Synthetic Primary Key)

2

name

Product name as it appears on the app

3

category

Product category like Fruits, Snacks, Beverages, etc.

4

mrp

Maximum Retail Price (originally in paise, converted to ₹)

5

discountPercent

Discount applied on MRP

6

discountedSellingPrice

Final price after discount (also converted to ₹)

7

availableQuantity

Units available in inventory

8

weightInGms

Product weight in grams

9

outOfStock

Boolean flag indicating stock availability

10

quantity

Number of units per package (mixed with grams for loose produce)

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Project Workflow

Database & Table Creation

Create SQL table with appropriate data types

Data Import

Load CSV using pgAdmin's import feature

Data Exploration

Count records, check null values, identify categories

Data Cleaning

Remove zero values, convert paise to rupees

Business Insights

Write queries for revenue and pricing analysis

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Database Setup

Table Structure

Created SQL table with appropriate data types including SERIAL PRIMARY KEY, VARCHAR fields for text, NUMERIC for pricing, and BOOLEAN for stock status.

```
CREATE TABLE zepto (
 sku_id SERIAL PRIMARY KEY,
 category VARCHAR(120),
 name VARCHAR(150) NOT NULL,
 mrp NUMERIC(8,2),
 discountPercent NUMERIC(5,2),
 availableQuantity INTEGER,
 discountedSellingPrice NUMERIC(8,2),
 weightInGms INTEGER,
 outOfStock BOOLEAN,
 quantity INTEGER
);
```

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Data Exploration & Cleaning

Exploration

- Counted total records
- Viewed sample data
- Checked for null values
- Identified distinct categories
- Compared in-stock vs out-of-stock

Cleaning

- Removed rows where MRP was zero
- Removed rows where discounted selling price was zero
- Converted mrp and discountedSellingPrice from paise to rupees

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Business Insights Queries

1. **Top 10 Best-Value Products**
   Find products with highest discount percentage

2. **High MRP Out-of-Stock Products**
   Identify expensive products currently unavailable

3. **Estimated Revenue by Category**
   Calculate total revenue potential per category

4. **Expensive Products with Minimal Discount**
   Filter MRP > ₹500 with low discount

5. **Top 5 Categories by Average Discount**
   Rank categories offering highest discounts

6. **Price Per Gram Analysis**
   Calculate value-for-money products

7. **Weight-Based Product Grouping**
   Group into Low, Medium, and Bulk categories

8. **Inventory Weight by Category**
   Measure total inventory weight per category

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Key Findings

Top Discount Products

Dukes Waffy wafers at 51% discount, RIO Mozzarella Block Cheese at 50% discount

Revenue Leaders

Health & Hygiene category leads with ₹64,180 estimated revenue

High-Value Out-of-Stock

Products with MRP > ₹300 currently unavailable

Value Products

Tata Salt at ₹24 for 1000g (₹0.02 per gram)

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